Imperial College London

Department of Electronic and Electrical Engineering

Examination 2025

Computational Sensing and Imaging

MSc and EEE/EIE PART IV

May, 2025 • Exam Duration: Three Hours

All Questions are Compulsory

Where unavoidable, make reasonable assumptions for your solutions.
Explain your assumptions in the answer script.

Markers: Ayush Bhandari and Sen Wang.

Questions [100 Marks]

Q1 Plenoptic Functions in Imaging.

(a) State the mathematical definition of the Plenoptic Function. **[1 Mark]**

The plenoptic function describes the intensity of each light ray in the world as a function of visual angle, wavelength, time, and viewing position.

(b) What is the dimensionality of the Plenoptic Function. **[1 Mark]**

It is a function of dimension-7.

(c) Identify the different variables of the function. **[2 Marks]**

Mathematically, it is described by $P = P(\mathbf{r}, \theta, \phi, \lambda, t)$.

(d) Map each parameter in the plenoptic function (a) to an imaging modality. **[3 Marks]**

$\mathbf{r}$, imaging at position $\mathbf{r}$; θ, ϕ, imaging angle; λ, imaging using light with wavelength λ (Hyper-spectral imaging); t, imaging at the time instance t (time-resolved imaging).

(e) Explain (mathematically) the creation of a 2D photograph from a plenoptic function. **[2 Marks]**

Assume monochromatic and time-invariant, the plenoptic function has 5 variables. The model is further reduced to 4D as the radiance along a ray remains constant throughout its length. Then, the 2D photography is produced by projecting 4D light field into a 2D plane as $I(s,t) = \iint P(u,v,s,t)dudv$.

Q2 Linear System Theory.

(a) Let $\mathcal{L}$ be a linear, time-invariant system.
Show that the response of this system to input signal $f(t)$ is given by,

$$f(t) \xrightarrow{\text{Input}} \boxed{\mathcal{L}} \xrightarrow{\text{Output}} \mathcal{L}[f](t) = (f * g)(t)$$

where $g(t) = \mathcal{L}[\delta](t)$ denotes the impulse response. **[4 Marks]**

Following the definition of convolution, we have

$$\mathcal{L}[f](t) = \mathcal{L}[f * \delta](t) = \int_{\mathbb{R}} f(t-\tau) \underbrace{\mathcal{L}[\delta](\tau)}_{=g(\tau)} d\tau = \int_{\mathbb{R}} f(t-\tau) g(\tau) d\tau = (f * g)(t).$$

(b) Show that $f_\omega(t) = e^{\jmath\omega t}$ is the eigenfunction for the system $\mathcal{L}$, satisfying that

$$f_\omega(t) \xrightarrow{\text{Input}} \boxed{\mathcal{L}} \xrightarrow{\text{Output}} \mathcal{L}[f_\omega](t) = \beta_\omega f_\omega(t).$$

Compute the corresponding eigenvalue β_ω. **[4 Marks]**

Following the definition, we have that

$$\mathcal{L}[f_\omega](t) = (f_\omega * g)(t) = \int_{\mathbb{R}} g(\tau) e^{\jmath\omega(t-\tau)} d\tau = e^{\jmath\omega t} \int_{\mathbb{R}} g(\tau) e^{-\jmath\omega\tau} d\tau = f_\omega(t)\, \widehat{g}(\omega)$$

where $\widehat{g}(\omega)$ is the Fourier transform of g at angular frequency ω.

Q3 Prony's Method for Frequency Estimation.

Consider sum-of-sinusoids (SoS) samples defined as

$$x\left[n\right]=\sum_{k=0}^{K-1}c_k e^{\jmath(\omega_k n+\varphi_k)},\quad n=0,\cdots,N-1. \tag{1}$$

Our goal is to estimate the sinusoidal parameters $\{c_k,\omega_k,\varphi_k\},\ k=0,\cdots,K-1$.

(a) Let $h\left[n\right],n=0,\cdots,K$ denote the filter with z-transform, $H(z)=\prod_{k=0}^{K-1}(1-e^{\jmath\omega_k}z^{-1})$. Show that,

$$\left(h*x\right)\left[n\right]=0.$$

[4 Marks]

Following the definition of convolution and sinusoidal samples x, we have

$$\begin{aligned}\left(h*x\right)\left[n\right]&=\sum_{m=0}^{K}h\left[m\right]x\left[n-m\right]=\sum_{m=0}^{K}h\left[m\right]\sum_{k=0}^{K-1}c_k e^{\jmath(\omega_k(n-m)+\varphi_k)}\\&=\sum_{k=0}^{K-1}c_k e^{\jmath(\omega_k n+\varphi_k)}\underbrace{\sum_{m=0}^{K}h\left[m\right]e^{-\jmath\omega_k m}}_{=H(e^{\jmath\omega_k})=0}=0.\end{aligned}$$

(b) Re-organize the annihilation equation $h\left[n\right],n=0,\cdots,K$ in vector-matrix form as $\mathbf{Ah}=\mathbf{0}$ and specify the dimension of $\mathbf{A}$. **[4 Marks]**

The annihilation equation can be rewritten in matrix form as

$$\underbrace{\begin{bmatrix}x\left[K\right]&x\left[K-1\right]&\cdots&x\left[0\right]\\x\left[K+1\right]&x\left[K\right]&\cdots&x\left[1\right]\\\vdots&\vdots&\ddots&\vdots\\x\left[N-1\right]&x\left[N-2\right]&\cdots&x\left[N-1-K\right]\end{bmatrix}}_{\mathbf{A}\in\mathbb{C}^{(N-K)\times(K+1)}}\underbrace{\begin{bmatrix}h\left[0\right]\\h\left[1\right]\\\vdots\\h\left[K\right]\end{bmatrix}}_{\mathbf{h}}=\mathbf{0}.$$

(c) To estimate $\mathbf{h}$, what is the condition on the number of samples N? **[1 Mark]**

N should satisfy $N\geqslant 2K$.

(d) Once $\mathbf{h}$ is known, how to determine $\{c_k,\omega_k,\varphi_k\},\ k=0,\cdots,K-1$? **[3 Marks]**

Once $\mathbf{h}$ is known, the frequencies can be found by computing the roots/zeros of the polynomial $H(z)$. The amplitudes c_k and phases φ_k can then be estimated via least-squares.

Q4 Lens Equation.

Let $\{r_1,r_2\}$ with $|r_1|<|r_2|$ denote the roots of the quadratic equation below

$$R^2-\beta R+\beta f(1-\rho)=0,\ \beta\neq 0.$$

Show that, $\{-r_1,r_2\}$ satisfy the Lens Maker's equation,

$$(\rho-1)\left(\frac{1}{R_1}-\frac{1}{R_2}\right)=\frac{1}{f}.$$

[4 Marks]

From Vieta's theorem, we know that the roots of the quadratic equation satisfy

$$r_1 + r_2 = \beta, \quad r_1 r_2 = \beta f(1-\rho) = 0. \tag{2}$$

Moreover, let $R_1 = -r_1$ and $R_2 = r_2$, then we have

$$-R_1 + R_2 = \beta, \quad -R_1 R_2 = \beta f(1-\rho) = 0. \tag{3}$$

Since $\beta \neq 0$, then

$$\frac{R_2 - R_1}{-R_1 R_2} = \frac{\beta}{\beta f(1-\rho)} \iff (\frac{1}{R_1} - \frac{1}{R_2}) = \frac{1}{f(\rho - 1)} \tag{4}$$

which leads to the desired result.

Q5 **Constrained Least Squares (CLS).**

Consider the following CLS problem:

$$\boxed{\min_{\mathbf{x}} \|\mathbf{WHx} - \mathbf{y}\|_2^2, \quad \text{such that} \quad \begin{cases} \mathbf{C_1 x} = \mathbf{b}_1 \\ \mathbf{C_2 x} = \mathbf{b_2} \end{cases}.} \tag{5}$$

(a) Write down the Lagrangian function. **[2 Marks]**

Define the Lagrangian,

$$\mathcal{L}(\mathbf{x}; \boldsymbol{\alpha}, \boldsymbol{\beta}) = \|\mathbf{WHx} - \mathbf{y}\|_2^2 + \boldsymbol{\alpha}^\top(\mathbf{C_1 x} - \mathbf{b}_1) + \boldsymbol{\beta}^\top(\mathbf{C_2 x} - \mathbf{b}_2).$$

(b) Compute the optimality conditions.

Hint: You can use: $\partial_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|_2^2 = 2\mathbf{A}^T(\mathbf{Ax} - \mathbf{b})$. **[2 Marks]**

The derivatives of the Lagrangian function is

$$\partial_{\mathbf{x}} \mathcal{L}(\mathbf{x}; \boldsymbol{\alpha}, \boldsymbol{\beta}) = 2\mathbf{WHx}^\top(\mathbf{WHx} - \mathbf{y}) + \mathbf{C_1}^\top \boldsymbol{\alpha} + \mathbf{C_2}^\top \boldsymbol{\beta}.$$

Combining the equality constraints, we have the optimality condition

$$\begin{aligned} 2\mathbf{WHx}^\top(\mathbf{WHx} - \mathbf{y}) + \mathbf{C_1}^\top \boldsymbol{\alpha} + \mathbf{C_2}^\top &= \mathbf{0} \\ \mathbf{C_1 x} &= \mathbf{b}_1 \\ \mathbf{C_2 x} &= \mathbf{b}_2. \end{aligned}$$

(c) Show that the Lagrangian multiplier satisfies that

$$\boldsymbol{\lambda} = 2\left(\mathbf{C}\left(\mathbf{A}^\top\mathbf{A}\right)^{-1}\mathbf{C}^\top\right)^{-1}\left(\mathbf{C}\left(\mathbf{A}^\top\mathbf{A}\right)^{-1}\mathbf{A}^\top\mathbf{y} - \mathbf{b}\right)$$

where $\mathbf{A} = \mathbf{WH}$, $\mathbf{C} = \begin{bmatrix} \mathbf{C}_1 \\ \mathbf{C}_2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \end{bmatrix}$. **[4 Marks]**

Simplifying the optimality condition with the auxiliary variable $\mathbf{A}, \mathbf{C}, \mathbf{b}$, we have

$$\begin{aligned} 2\mathbf{Ax}^\top(\mathbf{Ax} - \mathbf{y}) + \mathbf{C}^\top\boldsymbol{\lambda} &= \mathbf{0} \\ \mathbf{Cx} &= \mathbf{b}. \end{aligned}$$

From the first condition, we have

$$\mathbf{x} = (\mathbf{A}^\top \mathbf{A})^{-1}(\mathbf{A}^\top \mathbf{y} - 0.5\mathbf{C}^\top \boldsymbol{\lambda}).$$

Left-multiplying both sides with $\mathbf{C}$, we have

$$\mathbf{C}\mathbf{x} = \mathbf{C}(\mathbf{A}^\top \mathbf{A})^{-1}(\mathbf{A}^\top \mathbf{y} - 0.5\mathbf{C}^\top \boldsymbol{\lambda}) = \mathbf{b}$$

which results in the determination of the multiplier

$$\boldsymbol{\lambda} = 2\left(\mathbf{C}\left(\mathbf{A}^\top \mathbf{A}\right)^{-1}\mathbf{C}^\top\right)^{-1}\left(\mathbf{C}\left(\mathbf{A}^\top \mathbf{A}\right)^{-1}\mathbf{A}^\top \mathbf{y} - \mathbf{b}\right).$$

(d) Substituting the result of (c) in (b), show the optimal solution is:

$$\mathbf{x}^* = (\mathbf{A}^\top \mathbf{A})^{-1}\left(\mathbf{A}^\top \mathbf{y} - \mathbf{C}^\top(\mathbf{C}(\mathbf{A}^\top \mathbf{A})^{-1}\mathbf{C}^\top)^{-1}(\mathbf{C}(\mathbf{A}^\top \mathbf{A})^{-1}\mathbf{A}^\top \mathbf{y} - \mathbf{b})\right)$$

[2 Marks]

With $\boldsymbol{\lambda}$ determined, substituting $\boldsymbol{\lambda}$ into $2\mathbf{A}\mathbf{x}^\top(\mathbf{A}\mathbf{x} - \mathbf{y}) + \mathbf{C}^\top\boldsymbol{\lambda} = \mathbf{0}$ leads to the desired result.

Q6 Fluorescence Lifetime Imaging with Depth Sensors.

Consider a fluorescent sample with a lifetime parameter λ_0. The measurements resulting from the interaction of a continuous wave with the sample can be modeled as,

$$m(t) = \delta(t) * \rho_0 e^{-\frac{t}{\lambda_0}} u(t),$$

where m represents the measured signal, δ denotes the Dirac distribution, and u is the Heaviside step function. In applications such as microscopy, the sample is typically placed at a distance d_0 from the imaging plane. Consequently, the scene response function (SRF) can be expressed as

$$h_{\mathsf{FLI}}(t) = \underbrace{\Gamma_0 \delta(t - t_0)}_{\text{Direct Reflection}} + \underbrace{\rho_0 e^{-\frac{t-t_0}{\lambda_0}} u(t - t_0)}_{\text{Fluorescent Sample}}, \quad t_0 = 2\frac{d_0}{c}.$$

(a) What is the Fourier Transform of the scene response function $h_{\mathsf{FLI}}(t)$? **[4 Marks]**

The solution follows from Problem 1.b in Problem Sheet 2,

$$\widehat{h}_{\mathsf{FLI}}(\omega) = \left(\Gamma_0 + \frac{\rho_0 \lambda_0}{1 + \jmath\omega\lambda_0}\right) e^{-\jmath\omega t_0}.$$

(b) Consider the case of continuous-wave time-of-flight imaging, where the probing signal is defined as $p(t) = 1 + p_0 \cos(\omega t)$ and a lock-in sensor is employed.
Show that the measurements are given by

$$m(t) = \left|\widehat{h}_{\mathsf{FLI}}(0)\right| + \left|\widehat{h}_{\mathsf{FLI}}(\omega)\right| \frac{p_0^2}{2} \cos\left(\omega t - \angle\widehat{h}_{\mathsf{FLI}}(\omega)\right),$$

where $\widehat{h}_{\mathsf{FLI}}(\omega)$ is the Fourier Transform of the SRF computed in the above question.

[5 Marks]

Since sinusoids are eigen-functions of any linear, shift-invariant system say $h(t)$, we have that,

$$\cos(\omega t) * h(t) = \left|\widehat{h}(\omega)\right| \cos\left(\omega t + \angle\widehat{h}(\omega)\right)$$

and for the specific case when $p(t) = 1 + p_0\cos(\omega t)$, we have,

$$r(t) = p(t) * h_{\text{FLI}}(t) = \left|\widehat{h}_{\text{FLI}}(0)\right| + \left|\widehat{h}_{\text{FLI}}(\omega)\right| p_0 \cos\left(\omega t + \angle\widehat{h}_{\text{FLI}}(\omega)\right)$$

where the first term arises from the fact that the constant 1 in $p(t)$ can be interpreted as zero-frequency or, $p(t) = \cos(0t) + p_0\cos(\omega t)$. Again, following a procedure similar to the probem in Practice Sheet and computing,

$$C_{p,r}(t) = \lim_{T\to\infty} \frac{I}{2T} = \Gamma_0 + \frac{1}{2}\Gamma_0 p_0^2 \cos(\omega t + \theta),$$

we obtain

$$m(t) = \left|\widehat{h}_{\text{FLI}}(0)\right| + \left|\widehat{h}_{\text{FLI}}(\omega)\right| \frac{p_0^2}{2} \cos\left(\omega t - \angle\widehat{h}_{\text{FLI}}(\omega)\right).$$

Q7 (a) Consider the weighted regularization problem below:

$$\min_{\mathbf{x}} \|\mathbf{W}\mathbf{x} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{A}\mathbf{x}\|_2^2. \tag{6}$$

Compute the optimal solution.
Hint: You can use: $\partial_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{b}\|_2^2 = 2\mathbf{A}^T(\mathbf{A}\mathbf{x} - \mathbf{b})$. **[5 Marks]**

With positive weights, a more general form of the regularized objective function is given by:

$$J(\mathbf{x}) = \|\mathbf{y} - \mathbf{W}\mathbf{x}\|_2^2 + \lambda \|\mathbf{A}\mathbf{x}\|_2^2$$

where $\lambda > 0$. Taking the derivative, we get

$$\frac{\partial}{\partial \mathbf{x}} J(\mathbf{x}) = 2\mathbf{W}^\top(\mathbf{W}\mathbf{x} - \mathbf{y}) + 2\lambda\mathbf{A}^\top\mathbf{A}\mathbf{x}$$

Setting the derivative to zero,

$$\begin{aligned}\frac{\partial}{\partial \mathbf{x}} J(\mathbf{x}) = 0 &\implies \mathbf{W}^\top\mathbf{W}\mathbf{x} + \lambda\mathbf{A}^\top\mathbf{A}\mathbf{x} = \mathbf{H}^\top\mathbf{y} \\ &\implies (\mathbf{W}^\top\mathbf{W} + \lambda\mathbf{A}^\top\mathbf{A})\mathbf{x} = \mathbf{H}^\top\mathbf{y}\end{aligned}$$

So the solution is given by

$$\mathbf{x} = (\mathbf{W}^\top\mathbf{W} + \lambda\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{H}^\top\mathbf{y}$$

So,

$$\boxed{\min_{\mathbf{x}} \|\mathbf{y} - \mathbf{W}\mathbf{x}\|_2^2 + \lambda \|\mathbf{A}\mathbf{x}\|_2^2, \implies \mathbf{x} = (\mathbf{W}^\top\mathbf{W} + \lambda\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{H}^\top\mathbf{y}} \tag{13}$$

(b) In the frequency estimation problem in Eq. (1) of Q3, the measurements are usually noisy. Hence, the estimation of $\mathbf{h}$ in Q3 (b) is formulated as

$$\min_{\mathbf{h}} \|\mathbf{A}\mathbf{h}\|_2^2, \quad \text{subject to} \quad \|\mathbf{h}\|_2 = 1. \tag{14}$$

Utilizing the result in (a), show that the optimal solution to the minimization above is the eigenvector of $\mathbf{A}^\top\mathbf{A}$ associated with the minimal eigenvalue. **[5 Marks]**

Define the Lagrangian function,

$$\mathcal{L}(\mathbf{h};\lambda) = \frac{1}{2}\|\mathbf{A}\mathbf{h}\|_2^2 - \frac{\lambda}{2}(\|\mathbf{h}\|_2^2 - 1).$$

Computing the optimality condition leads to

$$\partial_{\mathbf{h}}\mathcal{L}(\mathbf{h};\lambda) = \mathbf{A}^\top\mathbf{A}\mathbf{h} - \lambda\mathbf{h} = \mathbf{0} \Longleftrightarrow \mathbf{A}^\top\mathbf{A}\mathbf{h} = \lambda\mathbf{h}. \tag{15}$$

Substituting the result above into the objective function, we have that

$$\|\mathbf{A}\mathbf{h}\|_2^2 = \mathbf{h}^\top\mathbf{A}^\top\mathbf{A}\mathbf{h} = \lambda\mathbf{h}^\top\mathbf{h} = \lambda.$$

Hence, the minimizer is the eigenvector of $\mathbf{A}^\top\mathbf{A}$ that corresponds to the minimal eigenvalue.

Q8 Sparse Recovery and Soft-Thresholding.

Consider the following recovery problem:

Let $\mathbf{x} \in \mathbb{R}^N$ be a sparse sequence with M nonzero elements. The observed measurements are characterized as $\mathbf{y} = \mathbf{A}\mathbf{x} + \mathbf{w}$ where $\mathbf{A}$ is certain orthogonal matrix and $\mathbf{w}$ is the vector of noise.

(a) Write down the least squares solution. **[1 Marks]**

The least square solution is $\mathbf{x} = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top\mathbf{y}$.

(b) Outline the issues of the least square solution. **[2 Marks]**

The issues lie in: (1) robustness to outliers and (2) promoting the sparsity.

(c) Write down the optimization formulation that leverages the sparsity of $\mathbf{x}$.
Hint: Use the ℓ_0-norm. **[2 Marks]**

Considering the sparse signal structure, the problem can be formulated as

$$\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2, \quad \text{subject to} \quad \|\mathbf{x}\|_{\ell_0} = M. \tag{16}$$

(d) Describe the main steps of the Basis Pursuit algorithm that solves (c). **[5 Marks]**

Let $\mathbf{A} = [\mathbf{a}_1\ \mathbf{a}_2\ \cdots \mathbf{a}_L] \in \mathbb{R}^{N\times L}$. (1) At the initialization stage, compute the inner product between the measurements and each column/atoms of $\mathbf{A}$:

$$d_i = \frac{\langle \mathbf{a}_i, \mathbf{y}\rangle}{\|\mathbf{a}_i\|_2^2}$$

which measures the contribution to the observed data. Remove the column with the maximal d_i from $\mathbf{A}$ and calculate the residue $y \leftarrow y - d_i\mathbf{a}_i$. Construct the new atom matrix with $\mathbf{a}_i$. (2) At l-th iteration, given the residue $y^{[l]} = y^{[l-1]} - d_i\mathbf{a}_i$, compute the inner product between the residue and remaining columns. Remove the most significant column from $\mathbf{A}$ and add it to the new atom matrix. Keep repeating the iterations until the number of columns of the new atom matrix reaches M.

(e) What are the potential issues of the optimization scheme in (c) and the approach in (d)? **[3 Marks]**

(1) Discontinuity and non-differentiability of the ℓ_0 norm. (2) Correlation requirement on the columns of $\mathbf{A}$.

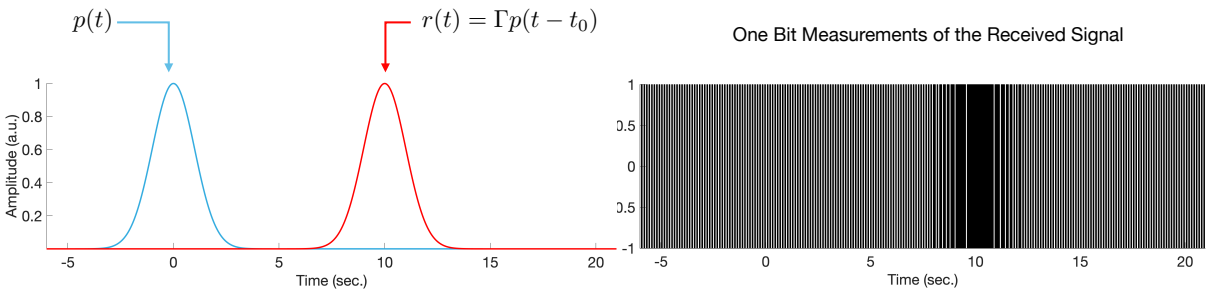


Fig. 1: Illustration of the one-bit time-resolved imaging.

(f) Due to the limitations, the problem formulation in (c) is generally relaxed. Using the regularization strategy, write down the relaxed optimization problem.
Hint: Replace the ℓ_0-norm in (c) with ℓ_1-norm. **[2 Marks]**

The relaxed regularization problem can be formulated as

$$\min_{\mathbf{x}} \|\mathbf{A}\mathbf{x} - \mathbf{y}\|_2^2 + \lambda \|\mathbf{x}\|_{\ell_1} \tag{17}$$

(g) Let $\mathbf{A} = \mathbf{Id}$, write down the solution to the minimization problem in (f).
Hint: You can use the soft–thresholding operator. **[4 Marks]**

Define the cost function:

$$\mathcal{L}_\lambda(\mathbf{x}) = \|\mathbf{y} - \mathbf{x}\|_2^2 + \lambda \|\mathbf{x}\|_{\ell_1} = \sum_{n=0}^{N-1} \left((y_n - x_n)^2 + \lambda |x_n| \right)$$

since $\mathbf{A} = \mathbf{Id}$. We compute the derivative of the cost function:

$$\frac{\partial}{\partial \mathbf{x}} \mathcal{L}_\lambda(\mathbf{x}) = 2(x_n - y_n) + \lambda \operatorname{sgn}(x_n)$$

where sgn is the sign (signum) function. To obtain the optimal value of $\mathbf{x}$, we set the derivative of the cost function to zero

$$\frac{\partial}{\partial \mathbf{x}} \mathcal{L}_\lambda(\mathbf{x}) = 0 \quad \Rightarrow \quad y_n = \left(x_n + \frac{\lambda}{2} \operatorname{sgn}(x_n) \right) \text{ (element-wise)}$$

To define x_n in terms of y_n, we use the inverse function method.

Q9 One-Bit Time-Resolved Imaging.

Given a bandlimited pulse $p(t)$ and its samples are $p[n] = p(nT)$ where T is the sampling period.

(a) Outline the difficulty of achieving mega-pixel resolution for existing time-resolved imaging. **[2 Marks]**

Time-resolved imaging preserves the time variable of the plenoptic function which captures a scene-dependent time profile at a specific time resolution, usually at the magnitude of nanoseconds. The temporal resolution is the number of pixels at the ToF sensor while the spatial resolution describes the time scale of exposure. Due to the time profile at each pixel, the data size explodes largely as the number of pixels increases or the time scale increases.

(b) Due to this limitation, we consider combining the one-bit sampling with time-resolved imaging. In one-bit sampling, the samples $q[n] \in \{-1, 1\}$ are given by,

$$\begin{cases} q[n] & = \operatorname{sgn}(u[n-1] + f[n]) \\ u[n] & = u[n-1] + f[n] - q[n] \end{cases}.$$

Based on the illustration in Fig. 1, write down the one-bit samples of the reflected signal $r(t)$ as a single equation in terms of p and u. **[2 Marks]**

The reflected measurements can be written as $r[n] = \Gamma p(nT - t_0)$. Combining with the one-bit sigma-delta modulation, we have

$$q[n] = r[n] - (u * v)[n] = \Gamma p(nT - t_0) - (u * v)[n]$$

where $v[n] = \delta[n] - \delta[n-1]$.

(c) From Q9(b) above, derive an equivalent expression in the Fourier domain. **[2 Marks]**

Let $\widehat{q}$ denote the Discrete Time Fourier Transform of q, then we have

$$\widehat{q}(m) = \Gamma e^{-\jmath\omega t_0}\widehat{p}(\omega) - \widehat{u}(\omega)\,\widehat{v}(\omega).$$

(d) To find the unknown time delay t_0, a common approach is to utilize *matched filtering*. In this way, one can estimate t_0 by solving the following maximization

$$t_0^\star = \arg\max_t |(r * \overline{p})(t)|, \quad \overline{p}(t) = p(-t). \tag{18}$$

Show that t_0 can be found by also utilizing match filtering for the one-bit samples q.

Hint: Utilize the fact that matched filtering with pulse p is equivalent to low-pass filtering.

[4 Marks]

Notice that, q contains two constituent components:

$$q[n] = \underbrace{\Gamma p(nT - t_0)}_{\text{Low-pass}} - \underbrace{(u * v)[n]}_{\text{High-pass}}.$$

Hence, performing matching filtering with p results in low-pass filtering. Since, $\widehat{v}$ is a high-pass filter. Hence,

$$(u * v * \overline{p})(t) \approx 0, \quad \forall t \in \mathbb{R}.$$

Then, equivalently we have,

$$(r * \overline{p})(t) \approx (p * \overline{p})(t), \quad \forall t \in \mathbb{R}.$$

This suggests that performing matching filtering to the one-bit measurements of the reflected signal r is as if it were being done for the original pulse p, which explains the claim.

Q10 Sparse-Deconvolution for Multi-Depth Imaging.

As illustrated in Fig. 2,

in multi-depth imaging, the measurements are given by

$$m(t) = \sum_{k=0}^{K-1} c_k \delta(t - t_k) * \varphi(t) = \sum_{k=0}^{K-1} c_k \varphi(t - t_k), \quad t = nT \tag{19}$$

where T is the sampling period and $\varphi(t)$ is the *known* Ω-bandlimited probing function.

Our goal is to find the scene parameters $\{c_k, t_k\}, k = 0, \cdots, K-1$.

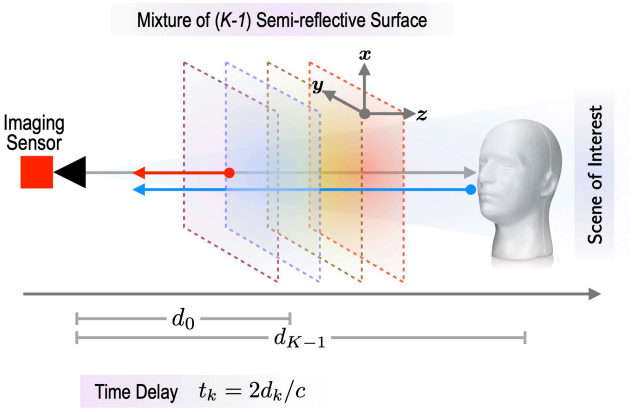


Fig. 2: Multi-Depth Imaging.

(a) Match filtering (see Q9 (d)) could be utilized to find the scene parameters. Describe the potential difficulty in estimating the scene using match filtering. **[2 Marks]**

Due to the smooth/bandlimited nature of the probing function, the auto-correlation function

$$f(\tau) = (m * \overline{\varphi})(t)$$

has a poor temporal resolution as its width/support expands because of the convolution operation.

(b) Write the Fourier transform of (19). **[2 Marks]**

As shown in class slides and the problem sets,

$$\widehat{m}(\omega) = \widehat{\varphi}(\omega) \sum_{k=0}^{K-1} c_k e^{-\jmath\omega t_k}.$$

(c) Show that the scene parameters can be found using the Prony's method (see (1)). **[3 Marks]**

Let $\widehat{m}$ denote the Fourier transform of m. Then, (19) in Fourier domain reads

$$\widehat{m}(\omega) = \widehat{\varphi}(\omega) \sum_{k=0}^{K-1} c_k e^{-\jmath\omega t_k}.$$

Since φ is known, we can move $\widehat{\varphi}$ to the LHS, resulting in

$$\frac{\widehat{m}(\omega)}{\widehat{\varphi}(\omega)} = \sum_{k=0}^{K-1} c_k e^{-\jmath\omega t_k}$$

which amounts to the frequency estimation problem.

This can be solved exactly using Prony's method previously used for (1).

Q11 What is the difference between RGB image and hyperspectral image? **[2 Marks]**

RGB Image: high spatial resolution with low spectral resolution; hyperspectral image: low spatial resolution and high spectral resolution.